



- 4 A program includes the following assignment statement:

`Result ← STR_TO_NUM(x) / STR_TO_NUM(y)`

When the program evaluates the expression in the statement, it performs a calculation.

Variable `Result` is of type real and variables `x` and `y` are of type string.

Two checks are required before the calculation is performed:

1. The two strings represent valid numeric values.
2. The numeric value of string `y` is not zero.

- (a) Identify the type of error that could occur if these checks are **not** carried out **and** state a cause of this error.

Type

.....

Cause

.....

[2]

- (b) The designer considers implementing the **checks and calculation** as a module (a procedure or a function). One reason for this is that the same checks and calculations are performed at several places in the program.

Give **another** reason why this is a suitable approach **and** state what is avoided by this approach.

Reason

.....

Avoided

.....

[2]

